

AR Lecture 3



Introduction to Augmented Reality

**Presented
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- **Definition of Augmented Reality**

Augmented Reality (AR) is a technology that **enhances the real-world environment by overlaying digital information** such as text, images, audio, or 3D objects in real time.

- **Key Characteristics of AR:**

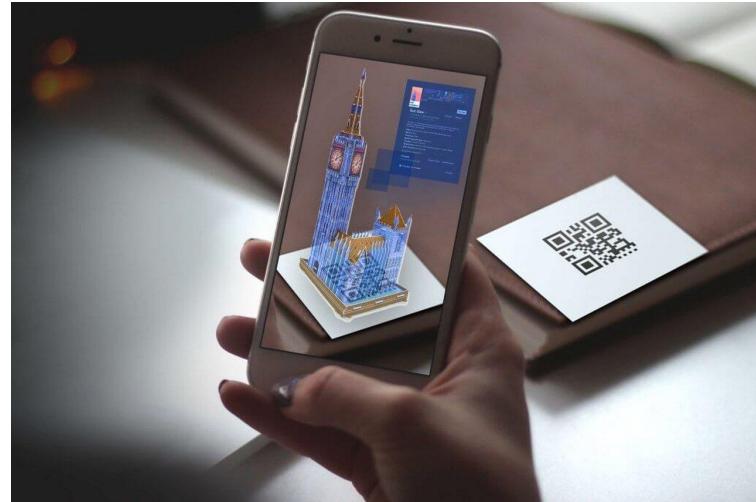
- Combines real and virtual content
- Real-time interaction
- Accurate 3D registration
- Works in real-world environments



Types of Augmented Reality

A. Marker-Based AR (10 minutes)

- **Description:**
Uses predefined visual markers (QR codes, black-and-white patterns) to trigger digital content.
- **Working Principle:**
Camera detects marker → identifies pattern → overlays virtual object.
- **Examples:**
AR textbooks, product packaging AR
- **Advantages:**
 - High accuracy
 - Easy implementation
- **Limitations:**
 - Requires visible marker
 - Limited flexibility



B. Marker-less AR (10 minutes)

- **Description:**
Does not require markers; uses environmental features.
- **Technologies Used:**
 - SLAM (Simultaneous Localization and Mapping)
 - Depth sensors
 - Computer vision
- **Examples:**
AR furniture placement apps, indoor navigation
- **Advantages:**
 - More natural interaction
 - Works in dynamic environments
- **Limitations:**
 - Computationally intensive
 - Requires powerful hardware



C. Location-Based AR (10 minutes)

- **Description:**
Uses geographical data to overlay virtual content based on user location.
- **Technologies Used:**
 - GPS
 - Compass
 - Accelerometer
- **Examples:**
Pokémon Go, location-based tourism apps
- **Advantages:**
 - Outdoor usability
 - Scalable
- **Limitations:**
 - GPS accuracy issues
 - Limited indoor usage



AR Pipeline

- **Definition:**
- The AR pipeline is a **sequence of processes** that enables digital content to be accurately placed over the real world.
- **Steps in AR Pipeline:**
- **Scene Capture** – Camera captures real-world view
- **Tracking & Registration** – Determines position and orientation
- **Processing & Rendering** – Generates virtual content
- **Display** – Combines real and virtual images

Tracking Methods in AR

Common Tracking Techniques:

- **Marker-based Tracking**
- **Feature-based Tracking**
- **Sensor-based Tracking**
- **Hybrid Tracking**

Recap & Interaction

Key Takeaways:

- AR enhances reality, not replaces it
- Different AR types serve different applications
- Tracking accuracy is critical in AR